

Abhishek Singh

B.Tech Electrical Engineering, IIT Patna (Batch of 2027)
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Education

Indian Institute of Technology Patna

B.Tech in Electrical Engineering

Relevant Coursework: DSA, AI & ML, Probability & Statistics, DSP, Signals & Systems, Linear Algebra

2023–2027

CPI: 6.41/10.0

Scholastic Achievements

- AIR 1046 in GATE (ECE), Score 662.
- AIR 5701 in JEE Advanced and AIR 4311 in JEE Main.
- Solved 1000+ DSA problems across competitive programming platforms.
- Completed extensive ML/DL coursework including Transformers, Computer Vision and AutoEncoders.

Research & Leadership Experience

Video Surveillance Research Intern, IIT Patna

Under Prof. Mahesh Kumar Kolekar

- Developed AutoEncoder-based anomaly detection systems for intelligent surveillance using PyTorch.
- Designed Hybrid Patch Feature Scoring for anomaly localization and scene-level anomaly detection.
- Trained and evaluated models on UCSD Pedestrian, CUHK Avenue and ShanghaiTech datasets.

Software Head, Inditronix AI Labs

- Led software architecture and development for healthcare AI solutions.
- Developed and deployed production infrastructure on AWS EC2.
- Built ESP32-based IoT applications and coordinated multidisciplinary software teams.

Projects

AI Surveillance System

- Developed AutoEncoder-based video anomaly detection framework using PyTorch.
- Designed Hybrid Patch Feature Scoring and evaluated on UCSD, CUHK Avenue and ShanghaiTech datasets.
- Built end-to-end training, inference and evaluation pipelines.

USAS Ingestor Engine

- Built scalable ingestion pipeline for multi-camera CCTV and DOT camera streams.
- Developed synchronization, preprocessing and storage modules for surveillance analytics.

Custom 3D Rendering Engine

- Developed custom real-time 3D rendering engine with gesture-based interaction.
- Built spline editor and procedural scene generation framework.

Competitive Programming

Codeforces (journey_to_grandmaster): 303+ solved, Rating 1241 (Pupil), Best Rank 2295 (Div-2)

CodeChef (abhishek_5701): 174+ solved, Rating 1643 (3-Star), Best Ranks 150 (Div-3), 459 (Div-2)

LeetCode: 121+ solved

1000+ DSA problems solved across online judges.

GeeksforGeeks: 183+ solved

Kaggle

Participated in ML competitions involving healthcare, finance and environmental prediction, including Problematic Internet Use Prediction, Loan Payback Prediction and Rainfall Dataset Prediction.

Technical Skills

Languages: Python, C++, C, SQL

ML/AI: TensorFlow, PyTorch, Deep Learning, Computer Vision, CNNs, Transformers, AutoEncoders, Anomaly Detection

Tools: Linux, Git, AWS EC2, OpenCV, NumPy, Pandas, MEEP, ESP32